

SUKKUR ROHRI, Pakistan

WMO: 417250

Lat: 27.667N

Lon: 68.900E

Elev: 223

StdP: 14.58

Time Zone: 5.00 (E05)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	46.1	48.2	28.6	22.8	69.4	33.3	28.1	67.0	9.2	61.3	7.6	61.7	1.0	50	0.374

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	21.5	113.2	76.0	110.8	76.6	108.6	77.8	84.6	101.4	83.9	100.5	83.3	99.8	2.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.8	161.4	93.3	79.9	156.3	93.5	79.1	152.1	93.8	49.4	102.1	48.5	101.4	47.7	100.8	94.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	9.9	7.8	6.7	40.8	118.3	4.9	1.6	37.3	119.4	34.4	120.4	31.7	121.3	28.1		
(5)				38.7	88.5	3.4	2.1	36.3	90.0	34.3	91.2	32.4	92.4	30.0	93.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.3	60.3	66.3	76.9	89.1	97.0	98.1	96.1	92.5	90.6	84.4	73.1	63.1
	DBStd	13.92	3.83	5.10	6.61	5.45	3.91	3.38	2.84	2.82	2.79	4.02	4.72	4.47
	HDD50	1	1	0	0	0	0	0	0	0	0	0	0	0
	HDD65	289	153	39	4	0	0	0	0	0	0	0	1	92
	CDD50	11805	318	455	833	1172	1456	1443	1428	1316	1218	1067	694	405
	CDD65	6619	6	75	372	722	991	993	963	851	768	602	245	31
	CDH74	99604	62	618	3912	10569	16548	16881	16010	13426	11712	7511	2089	266
	CDH80	66504	1	152	1917	6823	12115	12570	11547	8978	7431	4191	744	35
Wind	WSAvg	2.2	1.4	2.0	2.1	2.2	2.4	3.5	3.5	3.2	2.4	1.4	1.3	1.1
Precipitation	PrecAvg	4.6	0.2	0.3	0.2	0.3	0.2	0.4	1.5	1.1	0.4	0.1	0.0	0.2
	PrecMax	18.0	1.9	1.4	0.9	2.0	1.3	1.5	8.3	5.7	7.3	2.3	0.2	4.4
	PrecMin	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	3.8	0.4	0.4	0.3	0.5	0.3	0.5	2.2	1.6	1.5	0.4	0.0	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.0	88.7	103.8	112.1	116.8	116.5	110.8	105.6	105.6	102.6	94.0	84.3
		MCWB	60.6	65.7	70.6	72.4	74.5	76.3	82.7	81.8	79.4	75.7	69.2	65.9
	2%	DB	75.7	84.7	98.5	108.7	113.7	113.2	108.5	103.9	103.2	100.0	90.4	79.8
		MCWB	59.7	65.8	69.2	71.9	74.9	77.5	82.7	81.6	79.6	75.3	68.6	62.8
	5%	DB	72.9	80.9	94.1	105.6	111.5	110.5	106.6	102.3	101.1	97.7	87.5	76.8
		MCWB	58.1	63.9	68.8	71.8	74.6	78.8	82.6	81.5	79.3	74.3	68.1	61.0
	10%	DB	70.3	77.5	90.1	102.4	108.9	108.2	104.8	100.6	99.3	95.4	84.5	73.9
		MCWB	57.1	61.9	68.2	71.2	75.3	79.8	82.4	81.6	79.1	73.2	66.6	59.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.7	70.5	74.4	77.8	82.9	85.7	86.0	85.2	84.3	81.1	73.2	67.2
		MCDB	73.9	82.6	94.5	101.4	103.4	103.8	104.6	97.5	98.3	94.7	87.5	79.2
	2%	WB	61.6	68.4	72.4	75.9	81.4	84.1	84.9	84.1	83.0	79.6	71.5	65.1
		MCDB	72.4	80.0	91.3	99.5	102.9	103.7	103.1	97.5	96.7	93.4	85.5	76.2
	5%	WB	60.0	66.2	70.9	74.6	80.0	83.2	84.1	83.4	82.1	78.1	69.8	63.3
		MCDB	70.6	78.7	88.4	98.0	101.8	102.4	102.1	97.1	95.6	92.0	83.6	73.8
	10%	WB	58.4	64.0	69.5	73.3	78.7	82.3	83.4	82.7	81.4	76.5	68.3	61.3
		MCDB	68.1	75.1	86.2	96.4	100.6	101.3	101.1	96.6	94.8	90.2	81.1	71.6
Mean Daily Temperature Range	5% DB	MDBR	20.5	21.7	22.5	24.5	24.3	21.5	18.6	16.5	17.4	21.7	22.1	21.3
		MCDBR	23.9	25.5	27.9	28.9	27.0	24.7	20.9	18.9	20.1	23.3	24.2	24.3
		MCWBR	11.4	10.8	9.3	8.7	8.3	7.2	6.0	5.9	6.0	7.1	8.7	11.1
	5% WB	MCDBR	21.6	22.5	24.4	26.4	24.3	21.1	19.0	16.4	16.7	19.7	21.9	22.0
		MCWBR	11.8	10.6	9.3	9.5	9.6	7.2	6.6	6.3	6.1	7.5	9.0	11.9
Clear-Sky Solar Irradiance	taub	0.487	0.493	0.537	0.606	0.645	0.721	0.799	0.723	0.598	0.553	0.573	0.525	
	taud	1.864	1.891	1.801	1.694	1.648	1.560	1.481	1.598	1.787	1.780	1.700	1.783	
	Ebn at Noon	231	244	242	230	221	204	188	201	223	223	202	210	
	Edh at Noon	56	58	68	77	81	88	95	84	68	64	65	58	
All-Sky Solar Radiation	RadAvg	1214	1491	1861	2141	2289	2191	2033	1960	1913	1684	1319	1149	
	RadStd	67	94	82	54	41	64	85	99	81	65	58	52	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page